

ECE 315: Signals and Systems

Midterm Exam II Sample

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This sample is for self-study in preparation for Midterm Exam II; solutions will not be released. The exam covers Chapter 3 of **SS**, as well as lecture notes and homework from **March 2** to **April 10**. A summary of key concepts and related sample questions is provided below.

- Fourier series of continuous-time periodic signals: representation, convergence conditions, properties of Fourier series, Parseval's theorem, LTI system response to periodic signals
- Fourier series of discrete-time periodic signals: representation, properties of Fourier series, Parseval's theorem, LTI system response to periodic signals

1. (Fourier series representation) Consider the following continuous-time periodic signal:

- $x(t)$ is periodic with period 2 and $x(t) = \begin{cases} e^{-t} & \text{when } 0 \leq t \leq 1 \\ 1 & \text{when } 1 \leq t < 2 \end{cases}$

Compute its Fourier series representation, and select which types of convergence and justify your choice:

- ☐ Convergence everywhere
- ☐ Convergence with no energy difference
- ☐ Convergence except for discontinuities

2. (Fourier series representation) Consider the following Fourier series coefficients for each discrete-time periodic signal $x[n]$ with period 8:

- $a_k = \begin{cases} \sin(\frac{\pi}{3}k) & \text{when } 0 \leq k \leq 6 \\ 0 & \text{when } k = 7 \end{cases}$

Determine the signal $x[n]$.

3. (Fourier series representation, and its properties) Assume that we observe the following properties for a continuous-time signal $x(t)$:

- $x(t)$ is a real signal;
- $x(t)$ is periodic with period $T = 4$;
- $x(t)$ has Fourier series coefficients a_k that satisfy: (i) $a_k = 0$ for $|k| > 1$;
(ii) The signal with Fourier series coefficients $b_k = e^{-j\pi k/2} a_{-k}$ is odd;
- $\frac{1}{4} \int_2 |x(t)|^2 dt = \frac{1}{2}$.

Determine the signal $x(t)$.

4. (Discrete-time LTI system response to periodic signals) Consider a discrete-time linear-time invariant system with impulse response $h[n] = \frac{1}{2^n} u[n]$, and an input,

$$x[n] = \cos\left(\frac{\pi}{4}n\right) + \cos\left(\frac{3\pi}{4}n\right).$$

- (i) Determine the system function (system's eigenvalue) $H(z)$ and its frequency response $H(e^{jw})$;
- (ii) Compute the output of the LTI system for the input $x[n]$.